

Q1.

- (a) Prove by induction that, for all integers $n \geq 1$,

$$\sum_{r=1}^n r^3 = \frac{1}{4}n^2(n+1)^2$$

(4)

- (b) Hence show that

$$\sum_{r=1}^{2n} r(r-1)(r+1) = n(n+1)(2n-1)(2n+1)$$

(4)

(Total 8 marks)

Q2.

Given that $\mathbf{M} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$, prove that $\mathbf{M}^n = \begin{bmatrix} 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \\ 3^{n-1} & 3^{n-1} & 3^{n-1} \end{bmatrix}$ for all $n \in \mathbb{N}$

(Total 5 marks)

Q3.

Prove that $8^n - 7n + 6$ is divisible by 7 for all integers $n \geq 0$

(Total 5 marks)

Q4.

The polynomial $p(n)$ is given by $p(n) = (n-1)^3 + n^3 + (n+1)^3$.

- (a) (i) Show that $p(k+1) - p(k)$, where k is a positive integer, is a multiple of 9.

(3)

- (ii) Prove by induction that $p(n)$ is a multiple of 9 for all integers $n \geq 1$.

(4)

- (b) Using the result from part (a)(ii), show that $n(n^2 + 2)$ is a multiple of 3 for any positive integer n .

(2)

(Total 9 marks)

Q5.

The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 2, \quad u_{n+1} = \frac{5u_n - 3}{3u_n - 1}$$

Prove by induction that, for all integers $n \geq 1$,

$$u_n = \frac{3n+1}{3n-1}$$

(Total 6 marks)

Q6.

The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = \frac{3}{4} \quad u_{n+1} = \frac{3}{4 - u_n}$$

Prove by induction that, for all $n \geq 1$,

$$u_n = \frac{3^{n+1} - 3}{3^{n+1} - 1}$$

(Total 6 marks)

Q7.

(a) Prove by induction that, for all integers $n \geq 1$,

$$\frac{3}{1^2 \times 2^2} + \frac{5}{2^2 \times 3^2} + \frac{7}{3^2 \times 4^2} + \dots + \frac{2n+1}{n^2(n+1)^2} = 1 - \frac{1}{(n+1)^2}$$

(7)

(b) Find the smallest integer n for which the sum of the series differs from 1 by less than 10^{-5} .

(2)

(Total 9 marks)

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Q8.

(a) Given that

$$f(k) = 12^k + 2 \times 5^{k-1}$$

show that

$$f(k+1) - 5f(k) = a \times 12^k$$

where a is an integer.

(3)

(b) Prove by induction that $12^n + 2 \times 5^{n-1}$ is divisible by 7 for all integers $n \geq 1$.

(4)

(Total 7 marks)

Q9.

(a) Show that

$$(k+1)(4(k+1)^2 - 1) = 4k^3 + 12k^2 + 11k + 3$$

(2)

- (b) Prove by induction that, for all integers $n \geq 1$,

$$1^2 + 3^2 + 5^2 + \dots + (2n-1)^2 = \frac{1}{3}n(4n^2 - 1)$$

(6)

(Total 8 marks)

Q10.

The sequence $u_1, u_2, u_3, \dots$ is defined by

$$u_1 = 2, u_{k+1} = 2u_k + 1$$

- (a) Prove by induction that, for all $n \geq 1$,

$$u_n = 3 \times 2^{n-1} - 1$$

(5)

- (b) Show that

$$\sum_{r=1}^n u_r = u_{n+1} - (n+2)$$

(3)

(Total 8 marks)

Q11.

- (a) Show that $\frac{1}{(k+2)!} - \frac{k+1}{(k+3)!} = \frac{2}{(k+3)!}$

(2)

- (b) Prove by induction that, for all positive integers n ,

$$\sum_{r=1}^n \frac{r \times 2^r}{(r+2)!} = 1 - \frac{2^{n+1}}{(n+2)!}$$

(6)

(Total 8 marks)

Q12.

Prove by induction that

$$\frac{2 \times 1}{2 \times 3} + \frac{2^2 \times 2}{3 \times 4} + \frac{2^3 \times 3}{4 \times 5} + \dots + \frac{2^n \times n}{(n+1)(n+2)} = \frac{2^{n+1}}{n+2} - 1$$

for all integers $n \geq 1$.

(Total 7 marks)

Q13.

- (a) Prove by induction that, if n is a positive integer,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(5)

- (b) Hence, given that

$$z = \cos \theta + i \sin \theta$$

show that

$$z^n + \frac{1}{z^n} = 2 \cos n\theta$$

(3)

- (c) Given further that $z + \frac{1}{z} = \sqrt{2}$, find the value of

$$z^{10} + \frac{1}{z^{10}}$$

(4)

(Total 12 marks)

Q14.

Prove by induction that for all integers $n \geq 1$

$$\sum_{r=1}^n (r^2 + 1)(r!) = n(n+1)!$$

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(Total 7 marks)

Q15.

- (a) Explain why $n(n+1)$ is a multiple of 2 when n is an integer.

(1)

- (b) (i) Given that

$$f(n) = n(n^2 + 5)$$

show that $f(k+1) - f(k)$, where k is a positive integer, is a multiple of 6.

(4)

- (ii) Prove by induction that $f(n)$ is a multiple of 6 for all integers $n \geq 1$.

(4)

(Total 9 marks)

Q16.

- (a) Prove by induction that, if n is a positive integer,

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$$

(5)

(b) Find the value of $\left(\cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6$.

(2)

(c) Show that

$$(\cos \theta + i \sin \theta)(1 + \cos \theta - i \sin \theta) = 1 + \cos \theta + i \sin \theta$$

(3)

(d) Hence show that

$$\left(1 + \cos \frac{\pi}{6} + i \sin \frac{\pi}{6}\right)^6 + \left(1 + \cos \frac{\pi}{6} - i \sin \frac{\pi}{6}\right)^6 = 0$$

(4)

(Total 14 marks)

Q17.

(a) Show that

$$\left(1 - \frac{1}{(k+1)^2}\right) \times \frac{k+1}{2k} = \frac{k+2}{2(k+1)}$$

(3)

(b) Prove by induction that for all integers $n \geq 2$

$$\left(1 - \frac{1}{2^2}\right)\left(1 - \frac{1}{3^2}\right)\left(1 - \frac{1}{4^2}\right) \dots \left(1 - \frac{1}{n^2}\right) = \frac{n+1}{2n}$$

(4)

(Total 7 marks)

Q18.

(a) Prove by induction that

$$2 + (3 \times 2) + (4 \times 2^2) + \dots + (n+1) 2^{n-1} = n 2^n$$

for all integers $n \geq 1$.

(6)

(b) Show that

$$\sum_{r=n+1}^{2n} (r+1) 2^{r-1} = n 2^n (2^{n+1} - 1)$$

(3)

(Total 9 marks)

Q19.

- (a) The function f is given by

$$f(n) = 15^n - 8^{n-2}$$

Express

$$f(n+1) - 8f(n)$$

in the form $k \times 15^n$.

(4)

- (b) Prove by induction that $15^n - 8^{n-2}$ is a multiple of 7 for all integers $n \geq 2$.

(4)

(Total 8 marks)



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